

Landslide Monitoring System Implementing IOT using video camera

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Abstract—Landslide is a natural calamity which has devastating effects that can lead to multiple deaths and loss of property. Thus, landslide monitoring is of substantial importance in reducing the catastrophic effects of landslides. In this paper, we propose a landslide monitoring system based on Raspberry Pi implementing IOT using a video camera. It performs real-time analysis of the region, based on the video stream acquired by the camera and applies computer vision algorithms to detect landslide and notify stakeholders via Android app. Raspberry Pi being a low-cost device with low power demands can be installed in any region.

Index Terms—Real-time, landslide detection, video stream, motion detection, Android, Computer Vision.

I. INTRODUCTION

Landslide is one of the geological disasters which have been affecting the lives of people significantly. Every year, landslide takes away a lot of lives and causes sincere damage to property and the people. Therefore, in the regions prone to landslides, it is necessary to establish Early Warning Systems that can detect and inform the concerned people about the occurrence of a landslide. Many systems have been proposed and many have been developed to serve this purpose. Wireless Sensor Networks based system[1] is a network of wireless sensors that are able to detect any anomaly in the given area. Surface morphology, LIDAR[2] considers topographical images to perform modelling of the surface and detecting recent activity of landslides. Image Thresholding By Genetic Programming[3] makes use of satellite images to detect landslide activity in a region. Automatic Landslide Detection from remote sensing imagery using a scene classification method based on BoVW and pLSA[4] uses complex algorithms and k-NN classifier to detect landslide region and non-landslide region. Landslide Mapping Using Imagery Acquired By a Fixed-Wing UAV [5].

Digital video cameras have been used extensively for monitoring mine slope conditions and warning against rock falls [7] [8]. Most of these Landslide Monitoring Systems

require costly instrumentation and maintenance which makes it really difficult for developing nations and nations with small economy to afford such systems to monitor every landslide prone area. This shows the need of a low cost system that can detect landslide in real time and alarm the people in danger so that the necessary action can be taken.

We here propose a novel standalone computer based system which continuously monitors a given region by an inexpensive video camera and apply image processing techniques to detect Landslides and upload images on a web server and to notify stakeholders on their mobile device (Android app/SMS) to further verify the catastrophe.

IoT(Internet of Things) refers to communicating and actuating network of wireless sensing networks that seem to blend seamlessly in the environment around us. Sharing of this information across various platforms develops a Common Operating Picture(COP).[9] Incited by the adaptation of a variety of modern technologies, IoT has stepped out of its infancy and has become the the next revolutionary technology in transforming the Internet into a fully integrated Future Internet.

Similar to other IOT based systems, the system proposed in this research (Figure 1) uses a camera to sense the surroundings to update relevant information to the server. The camera captures frames and based on the previous data, moving objects are detected. Analogous to the theory of center of mass in newtonian mechanics our system considers the center of mass of moving "pixels" to detect and estimate the direction of motion. Although this is a primitive method for motion detection and estimation, it proves to be a faster methodology to implement on a low cost computer like raspberry pi for real time notification.

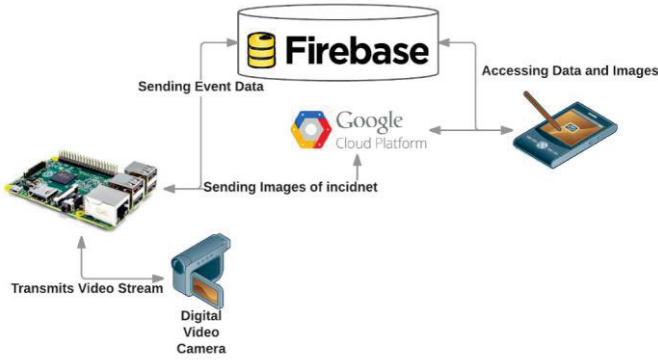


Fig.1. System Overview

The principal requirement for designing any system is the availability of sufficient test cases. Landslides being a natural phenomenon that generally occurs in remote regions, thus the videos of landslides are not available with us for research purposes. So to develop the above said system we have modeled our landslide test case against a waterfall due to opening of gates in a dam. The waterfall can be approximated to a layered landslide and thus, a landslide detection mechanism based on it has been developed.

The flow of the paper is as follows: Section 2 gives a brief of the proposed system. Finally Section 3 and Section 4 provides the results and conclusions respectively.

II. METHODOLOGY

A. Video Capture

The system takes input in the form of a video stream. Digital Video camera devices(not still cameras) are used. The Digital Video Camera Device then would capture live feed of a designated area from a far off region. An open source library called OpenCV is used to process the video from the device on the single board computer (Raspberry Pi).

B. Frame Capture and background Subtraction

While the video is being captured, the raspberry pi performs background subtraction on the captured frames. The background subtraction is performed using *Gaussian Mixture - based Background / Foreground Segmentation Algorithm*. The Gaussian mixture based approach as quoted in [10] works by modeling each pixel as a Gaussian mixture of neighbouring pixels instead of using one particular type of distribution for modelling all pixels explicitly. Based on these Gaussians, a classification for background or foreground is done. A pixel is not considered to belong to foreground until it falls under a foreground classified gaussian with sufficient and consistent evidence supporting it.

C. Removal of Noise

Proposed system uses a non-linear method for removal of noise referred to as median filtering. This method removes noise effectively while preserving the edges. In this method, a window traverses the entire image while calculating the median intensity value for each of the pixels in the image. The output intensity value of the pixel being processed is calculated using the intensity values of the neighbouring

pixels(or the pixels contained in the window) by sorting all these intensity values in numerical order and selecting the middle(median) value as the output intensity value for the pixel being processed.

D. Calculating Moment and Centroid

Image moment of an image is referred to as a particular weighted average (moment) of the image's pixel intensities used for interpretation. Image moments serve a great purpose describing objects post segmentation, calculating properties of image, namely its area (total intensity), centroid and information regarding its orientation. Our system uses moment to calculate centroid on the resultant frame obtained from the above operations. For a 2D continuous function $f(x,y)$ the moment (often called "raw moment") of the order $(p + q)$ is defined as,

$$M_{pq} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x^p y^q f(x, y) dx dy$$

for $p, q = 0, 1, 2, \dots$. Adapting this to scalar (grayscale) image with pixel intensities $I(x,y)$, raw image moments M_{ij} are thus calculated by

$$M_{ij} = \sum_x \sum_y x^i y^j I(x, y)$$

$$\text{Centroid: } \{x, y\} = \{M_{10}/M_{00}, M_{01}/M_{00}\}$$

Whenever a motion is detected, centroid is calculated and an array of 2×3 is updated with the coordinates of centroid. (Figure 3)

E. Landslide Detection

We form an array denoted named Centroid_positions[2][3] where N denotes the number of frames taken into consideration for motion detection, then values of centroid_position are updated as follows,

- Centroid_position[:,1] contains centroid coordinates when motion is detected that is, at the First frame of motion detection.
- Centroid_position[:,2] contains centroid coordinates when motion has been detected for N/2 consecutive frames.
- Centroid_position[:,3] contains centroid coordinates when motion has been detected for N consecutive frames.

After N consecutive frames (25 in our test case) detects motion, assuming the left top corner of image frame has (0,0) as coordinate and increases in left and downward direction, the gradient of points traced by centroid is calculated.

If the change in $\Delta y > 0$, then the net motion of landslide is said to be in downward direction. Further if $\Delta x > 0$, the landslide is interpreted to occur from left to right direction downhill and if $\Delta x < 0$, the landslide is interpreted to occur from right to left direction downhill.

F. Updation of Database and Sending Images

After the detection of motion for N consecutive frames and classification of it as a landslide, the three frames corresponding to the landslide (1st,(N/2)th and Nth frames respectively) are transmitted with a timestamp via raspberry pi to the database. A mobile application fetches this data from the database and shows it in an interactive way to the user of the app. Screenshots of the app are provided in Figure 4 and 5. Notifications are also sent to the authorized users so that necessary action could be taken to cope up with the situation.

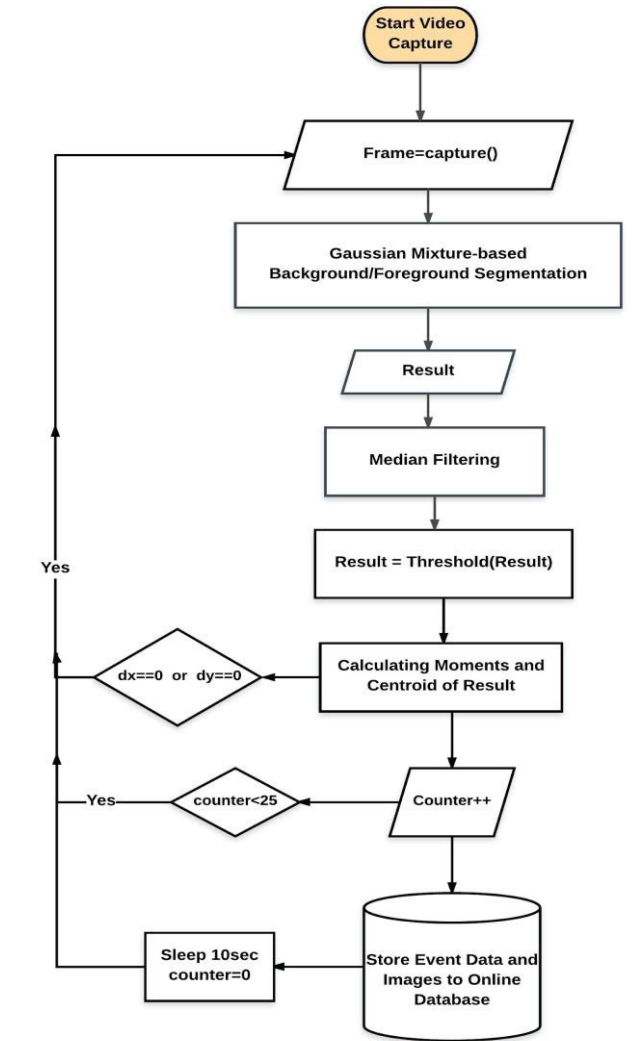


Fig.2. Process Overview

III. RESULTS

The proposed system was tested on a recorded footage of opening of gates of a water dam which can be considered as an appropriate model of a layered landslide, which provided viable results in our experiments. With a stable internet the database was updated in 3 seconds of the occurrence of the landslide. The mobile app was able to show the images of the landslide 5 seconds after its occurrence. Images of the landslide with the event data is shown by the mobile app in



Fig.3. Showing Movement of Centroid and Result after MOG and Median Filtering

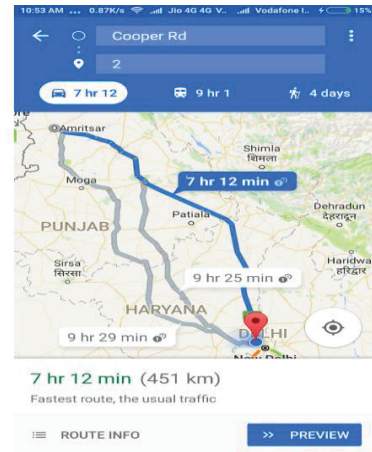


Fig.4. Screenshot of Navigation Feature



Fig.5. Screenshot of App showing Landslides

IV. CONCLUSION AND FUTURE WORK

In this paper, we proposed an innovative and economical system of landslide monitoring which would notify the concerned users. We have implemented real-time landslide detection and within 5-8 seconds, information regarding the incident reached the stakeholders of the mobile application. In future, we propose to further improve the method of landslide detection by implementing deep learning to detect such anomalies.

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